DESIGN AND IMPLEMENTATION OF AN

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BY

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DEPARTMENT OF COMPUTER SCIENCE

FACULTY OF COMPUTING

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CERTIFICATION

I hereby certify that this project work was carried out by **IDEHEN DESTINY** with the Matriculation Number **CMP2203517**, and the work is adequate in scope and quality for the partial fulfilment of the requirements for the award of a Bachelor of Science (B.Sc) degree in COMPUTER SCIENCE, University Of Delta, Agbor.

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(Project Supervisor)

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Dedication

This project is dedicated to God Almighty for His guidance, wisdom, and strength throughout the course of this work. I also dedicate this work to my parents and family for their constant prayers, love, and support, and to my lecturers and friends whose encouragement and guidance made the successful completion of this project.

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ABSTRACT

Having quick access to reliable information about healthcare facilities is very important, especially during emergency situations where every minute counts. In many developing countries such as Nigeria, people often face challenges when trying to locate nearby hospitals or determine the fastest route to reach them. This is mainly due to the lack of a centralized system that provides accurate and up-to-date hospital location information. As a result, delays in accessing medical care may occur, which can negatively affect patient outcomes.

This project presents the design and implementation of an **Online Hospital Location System using Google Maps**. The system is developed as a web-based application that allows users to easily search for nearby hospitals, view their locations on an interactive map, and obtain real-time directions from their current position. The Google Maps Application Programming Interface (API) is integrated to ensure accurate geolocation, mapping, and navigation services. In addition, the system provides essential hospital details such as names, addresses, and contact information to assist users in making informed decisions.

The proposed system is aimed at improving access to healthcare services, reducing response time during emergencies, and encouraging the effective use of web and geographic information technologies in the healthcare sector. The study demonstrates how location-based services and mapping technologies can be applied to address real-life challenges in healthcare delivery and enhance public safety.

CHAPTER ONE

INTRODUCTION

1.1 Background to the Study

Today's world is a world that revolves highly on the importance of technology for all aspects of human life, especially in the health sector. One of the most important affairs of human life is how human beings treat one another. The biggest challenges people encounter, especially in developing countries like Nigeria, is how important the location of hospitals is, for example, in emergency cases, many people and also the patient may not be aware of what service is offered at the nearest hospital to him or her, or the fastest route to get there.

In emergency cases, time is very important. Delays arising from the absence of the needed information will result in severe ramifications because of information, poor directions, or lack of familiarity with an area can lead to serious consequences. While there are existing general maps applications available, they are not necessarily focusing on hospitals, their services, or their availability in a way for patients, something simple and specific.

This project proposes Its objective is to facilitate the design and implementation of an online hospital. a location system that utilizes Google Maps for easier access in or near hospitals, their location, and directions. This system will serve as a web-based platform through which hospitals can be displayed on a map and The users can also search for their entries based on their location.

1.2 Statement of the Problem (Motivation)

With as many hospitals as are found in most regions, patients experience difficulties in locating the nearest or most relevant hospital in an event of a need to do so. This is due to the lack of a unified online system for hospital locations information. The lack of such a platform creates a lot of inefficiencies and waste of time, in cases where medical emergencies are concerned. Such delays may cost the deterioration of health conditions or even loss of life.

In addition, most of the current sources of information about hospitals are poorly organized, outdated, or inaccessible with ease by people unfamiliar with the locality. This fact makes the need

for an accurate, easily usable, and updated system of locating hospitals imperative. This work will seek to develop a web-based system applying Google Maps technology in order to provide proper location and routing data to hospitals accurately.

1.3 Aim and Objectives

Aim of the Study:

The purpose of this research work is to develop and implement an Online Hospital Location System using Google Maps to help users locate nearby hospitals and get directions to reach them.

Objectives of the Study:

The specific objectives of the study are to:

1. Design a web-based application that allows users to search for nearby hospitals.
2. Integrate Google Maps API to display hospital locations and provide navigation routes.
3. Provide accurate and up-to-date hospital information, including name, address, and contact details.
4. Enable users to obtain real time directions from their current location to selected hospitals.
5. Improve response time and decision-making during medical emergencies.

1.4 Significance of the Study

The relevance of this research work is based on its ability to enhance access to healthcare services and improve public safety for individuals. The proposed system will help patients, caregivers, and emergency services locate hospitals quickly and efficiently, while preventing delays in accessing healthcare. The research work encourages the effective use of technology in the healthcare sector.

The research work is an academic contribution to adding knowledge in the realms of web based systems, geographic information systems, and location based services. This research paper would

be helpful as a reference material for students and researchers who are interested in developing applications related to the health sector using mapping technologies.

In terms of academical contributions, it could be said that the present study will be making a significant contribution to the pool of knowledge, particularly with regards to systems derived from the internet, geographic systems, location systems, and so forth. Furthermore, it could be added that the present study will be used, particularly, as an aid for students and researchers who have interests with regards to exploring the internet and its application for health care purposes.

Moreover, it fosters efficient usage of information and communication technology, particularly in health institutions. For instance, with the incorporation of Google Maps into this website, with further location based functionality embedded, this study illustrates how technology could become effectively utilized to address specific health issues in the health sector, thereby fostering effective decision making processes.

1.5 Scope of the Study

The scope of this research work is restricted to the development of an Online Hospital Location System using Google Maps for the display of hospital locations and navigation services. The proposed system will be limited to hospitals in a specific geographical area and will include basic information about the hospitals and routing facilities. The advanced features of online appointment scheduling, ambulance dispatch, real time hospital capacity, and patient medical records are not included in the scope of this research work.

1.6 Operational Definition of Terms

- **Online Hospital Location System:** A web-based application designed to help users locate nearby hospitals and obtain directions using digital maps.
- **Google Maps API:** An application programming interface that allows developers to embed Google Maps features into web applications.

- **Location-Based Services (LBS):** Services that provide information or functionality based on a user's geographic location.
- **Geographic Information System (GIS):** A system designed to capture, store, analyze, and display geographically referenced data.
- **Web Application:** A software application accessed through a web browser over the internet.
- **Geolocation:** The process of determining the real-time geographic location of a user or device.

1.7 Organization of the Other Chapters

This project report is organized into five chapters.

Chapter One presents the introduction, including the background to the study, problem statement, aim and objectives, significance, scope, operational definitions, and organization of the chapters.

Chapter Two reviews relevant literature related to hospital location systems, location-based services, and Google Maps technology.

Chapter Three focuses on the system analysis and design, including the system architecture and design methodology.

Chapter Four discusses the system implementation, testing, and results.

Chapter Five presents the summary of findings, conclusion, and recommendations for future research and system improvement.

CHAPTER TWO

LITERATURE REVIEW

2.1 Conceptual Framework

Conceptual Framework

Several digital mapping systems and health related applications already exist. These may include Google Maps, Healthgrades, hospitals, amongst others. These may help the user locate the hospitals and navigate the direction to the hospitals. Most of these are either too general, mainly directing the user, or only available for certain locations in the country. For instance, some EHRs may be too general and mainly focus on the patients, medical appointments, and billing systems, without placing too much emphasis on the location and navigation for the hospitals and such systems. Most of the available options may not assist the user sufficiently and may only be available for the purpose and consistency needed for the user to find the hospitals, especially in the less developed areas within the country. This project extends the concept and the use of the online mapping system. There is the use of Google Maps for the ease and addressing the needs for the concept.

A conceptual framework, on the other hand, refers to "a way of showing how the research variables are interrelated and how these research variables interrelate to research study objectives." For instance, in the research study, the research variables are technology by location, Google Map API, hospital info, and accessibility.

The independent variables include location based services such as geolocation, mapping, and routing capabilities offered by Google Maps. The technology allows the system to locate the user's current position and display nearby hospitals on a digital map. The dependent variable is enhanced accessibility to healthcare facilities, which is determined by the ability of the user to quickly access hospitals and get correct directions.

The mediator variable has to do with the internet accessibility, the correctness of the information related to the hospital, and the compatibility with the device. This variable has to do with the efficiency of the system. A number of benefits may be derived from the conceptual framework.

Such benefits include that the technology provided by Google Maps helps one to locate hospitals faster. Secondly, correct information leads to faster responses with the use of the internet.

The significance of the research study lies in its potency towards improving the ease of providing health care services and promoting safety. the significance of the research study presented in this research proposal emanates from its aim to develop an Online Hospital Location System towards providing an efficient service for locating hospitals for the ease, comfort, and benefits of patients and caregivers. Hence, by providing exact and efficient health care services, safety is greatly promoted because health care services are provided in a timely manner. The aspect of health care being provided in a timely manner is very critical, especially for matters dealing with emergencies.

From the academic arena, it may be added to the array of available information that may be acquired about web system technology, Geographic Information System technology, and location based services. It can play the role of additional reference material for students working toward additional education in the fields, with the intention and goal of developing the technology for use in the arena of creating applications for the betterment and implementation of the health sector. It can also be used to aid further research for the development and establishment of additional efficient and sophisticated information systems for the betterment and improvement of the medical arena.

Moreover, the research embraces the need to harness information communication technology as an enhancement, particularly relating to solving issues in the health sector. Within this context, harnessing Google Maps through easy interaction of users with location based services through an application on the web serves to illustrate how technology can be used to address issues in the health sector, leading to innovation, as well as facilitating effective decision making. (Yasmin, 2023)

2.2 Theoretical Framework

The theory underpinning the research, Geographic Information System (GIS), and the Technology Acceptance Model (TAM), serve as the anchor for this research work

GIS theory helps in explaining how spatial data can be gathered, stored, analyzed, and visualized for effective decision making. In relation to the project, the application of GIS concepts is realized in Google Maps, which is used for the geographical depiction of the position of hospitals and for directions. This concept helps in implementing the idea that graphics work well in relation to the clear depiction of space.

The Technology Acceptance Model (TAM) refers to the process through which users can adopt technology based on their perceptions about the usefulness and ease of use of the technology. The Proposed Online Hospital Location System will be made convenient and effective to make users adopt it as an effective means to locate hospitals. According to TAM theory, when users perceive an online system to be useful and easy to use, they will be able to use it on a regular basis to accomplish the project goals. Algarni, A. (2018)

2.3 Review of Related Works

A number of researches have been conducted to determine the application of location-based systems in healthcare service delivery. An hospital locator application has been developed based on GIS and mobile technology to enhance the availability of health services in healthcare organizations. Researches clearly reveal that the use of digital maps in healthcare systems considerably reduces the search time to healthcare organizations, particularly in urban areas.

Other works also relate to the emergency response systems that employ GPS and map technology for assisting patients and emergency personnel. Additionally, the existing works include systems that lack up-to-date hospital data, require the use of specific platforms, or have intricate designs.

Previous studies have also shown how useful Google Maps API is for web applications when it comes to mapping accuracy and scalability. Though most applications have been developed for

mobile devices, not much has been mentioned on web-based applications for finding hospitals using a standard web browser.

The project relies on past studies and introduces a web-based system for finding hospitals using simple and accurate mapping applications. Abdelkarim, A. (2019)

There has been an implementation of a hospital locator as an application created through GIS technology along side mobile technology in a bid to enhance the availability of health services within a healthcare organization. The relevance of using digital maps within healthcare management to significantly minimize the time to seek healthcare services within an urban setting has been clarified by various research studies (Accessibility analysis of healthcare facility, 2017).

In addition to this, the same works apply to the response systems used in emergencies involving GPS technology and mapping to offer aid to patients and respondents during emergencies. Most important of all, the works exist, but none of them had any updates on hospitals. In addition to this, some of the works had complex designs compared to the current technology (Alexander et al., 2021).

Previous studies have also shown how useful Google Maps API is for web applications when it comes to mapping accuracy, and how web mapping APIs are used to estimate travel times to health facilities (Zheng et al., 2019; Gligorić et al., 2023). Though most applications have been developed for mobile devices, not much has been mentioned on web-based applications for finding hospitals using a standard web browser.

The project relies on past studies and introduces a web-based system for finding hospitals using simple and accurate mapping applications (JETIR, 2025).

2.4 GOOGLE MAPS TECHNOLOGY

Smartphones today made breakthroughs in processing power, higher random access memory, and secondary storage along with the number of extra features like internet access have opened doors to a broad range of applications development. An Android operating system (AOS) commonly used by Smartphone manufactures because it is an open source operating system based on the Linux kernel and designed primarily for touch screen mobile devices such as Smartphones and tablet computers. The internal hardware of Smartphones especially sensors like proximity, accelerometer and gyroscope sensors are used by some applications to respond to additional user actions (Akash, 2014).

2.5 Summary of Literature Review

From above literature reviews, it is established that in an effective health delivery system, for example, it is very important to have information regarding the locations of hospitals. From literatures regarding GIS and LBS technologies, mapping technologies have also been suggested to boost the abilities to find health facilities as well as reach them punctually (Boulos, 2003 Shaw & McGuire, 2014). The application of Google Maps API was also identified in literature reviews above to provide geolocation and navigation services as it is an accurate and attractive technology (Zheng et al. 2019 Gligorić et al. 2023).

Thirdly, the review indicates the need for a hospital finder system in the sense that, although a number of hospital finder systems exist, many of them face the challenge of a lack of usability, outdating, or poor accessible platforms (Alexander et al., 2021). The need for a hospital location system based on the various theories concerning the various location systems for a hospital was highlighted in the literature review. This leads us, through the course of the study, to the creation of the Online Hospital Location System, which takes advantage of the opportunities presented by Google Maps.

2.6 Conclusion

This study aimed to develop an online system to locate hospitals using Google Maps to improve their accessibility. This study was motivated by the problems of locating hospitals during emergency situations. There are limitations of time during emergency conditions; nonetheless, it would help to make hospital accessibility better as it is going to integrate Google Maps API.

Considering the findings of the research conducted through the literature search as well as the implementation of the system, it is safe to deduce the following regarding geographic information systems as well as web technologies: the efficiency of hospital navigation may be improved. There also exist limitations to the efficiency of the hospital finders developed thus far, which have in turn been addressed through the provision of information access to the users through the use of web technologies to access the proposed system via normal web browser usage.

In conclusion, Online Hospital Location System is a notable case proving the efficiency of employing various mapping techniques to tackle pressing issues facing the healthcare sector. The proposed solution is greatly helpful in enhancing the level of healthcare accessibility, facilitating time-saving during emergency response, and promoting the adoption of technology in the healthcare domain. This study contributes to the development of academic knowledge regarding web-based technology, GIS technology, and various location-based technology concepts.